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3. Boundary conditions for the hanging cable

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On the previous page, we found a differential equation for the shape of a hanging cable to be

$$y'' = \frac{1}{A} \sqrt{1 + (y')^2},$$

where A is determined by the mass per unit length of the cable, the horizontal tension in the zipline cable, and gravity.

Squaring both sides, this is

$$(y'')^2 = \frac{1}{A^2} (1 + (y')^2).$$

Differentiating implicitly this becomes

$$2y''y''' = \frac{1}{A^2} (2y'y'').$$

This equation is true if

$$y'' = 0 \quad \text{or} \quad y''' = \frac{1}{A^2} y'.$$

The differential equation $y'' = 0$ is solved by the straight line $y = ax + b$. This is the solution in the case that the length of cable is exactly equal to the distance between the end points. In all other cases, we must solve the differential equation $y''' = \frac{1}{A^2} y'$.

This is a third order differential equation, but we may solve it by setting $u = y'$ and solving the associated second order differential equation. Check for yourself that the differential equation

$$u'' = \frac{1}{A^2} u$$

is solved by a hyperbolic sinusoid of the form

$$u = \sinh\left(\frac{x - c_1}{A}\right).$$

Let us rename A to be the constant c_2 , to make it clear that this function has two free parameters:

$$u = \sinh\left(\frac{x - c_1}{c_2}\right).$$

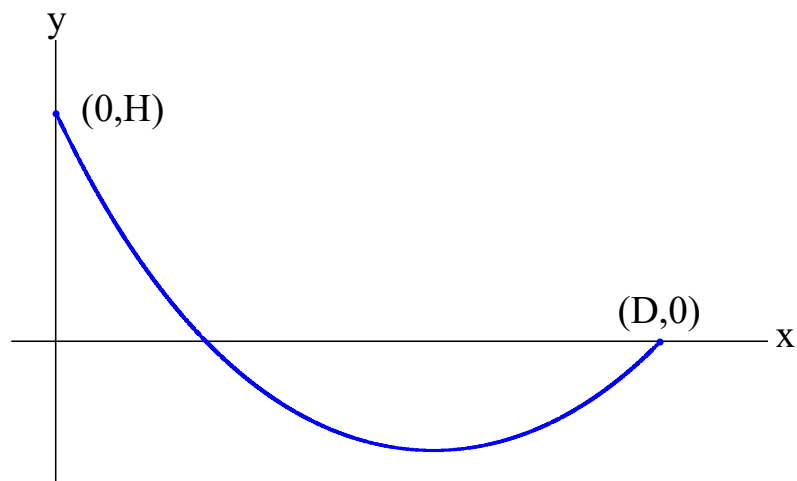
Therefore if $y' = u$, then y is a generalized hyperbolic cosine

$$y = c_3 + c_2 \cosh\left(\frac{x - c_1}{c_2}\right),$$

where c_1 , c_2 , and c_3 must be determined by initial conditions (or boundary conditions).

In order to solve for the unknown constants c_3 , c_2 , and c_1 , there are three conditions that must be satisfied:

- 1. The catenary must pass through the point (0,H).
- 2. The catenary must pass through the point (D,0).
- 3. The arc length of the catenary is L .



These conditions are called boundary conditions. Unlike initial conditions, these conditions do not uniquely specify a catenary. In this case, these boundary conditions specify two catenaries, one concave up and one concave down. Thus we can specify the catenary by forcing ourselves to choose the catenary that is concave up.

The boundary conditions give us a system of equations in terms of the vector of parameters $\mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$:

$$y(\mathbf{c}, 0) = H, \quad y(\mathbf{c}, D) = 0, \quad \text{length}(y(\mathbf{c})) = L.$$

In order for MATLAB to solve for $\mathbf{c}$ using **fsolve** (see documentation [here](#)), we should rewrite these three equations in the form

$$F_1(\mathbf{c}) = 0, \quad F_2(\mathbf{c}) = 0, \quad F_3(\mathbf{c}) = 0.$$

We ask you to specify the first two equations in the MATLAB code below.

Solving for the boundary conditions (External resource) (1.0 points possible)

```
26 yPlot = cOut(3) + cOut(2)*cosh((xPlot-cOut(1))/cOut(2));
27 plot(xPlot,yPlot,'r','linewidth',2,'markersize',10);
28 set(gcf, 'Position', [10 10 1000 1000]);
```

▶ Run Script ?

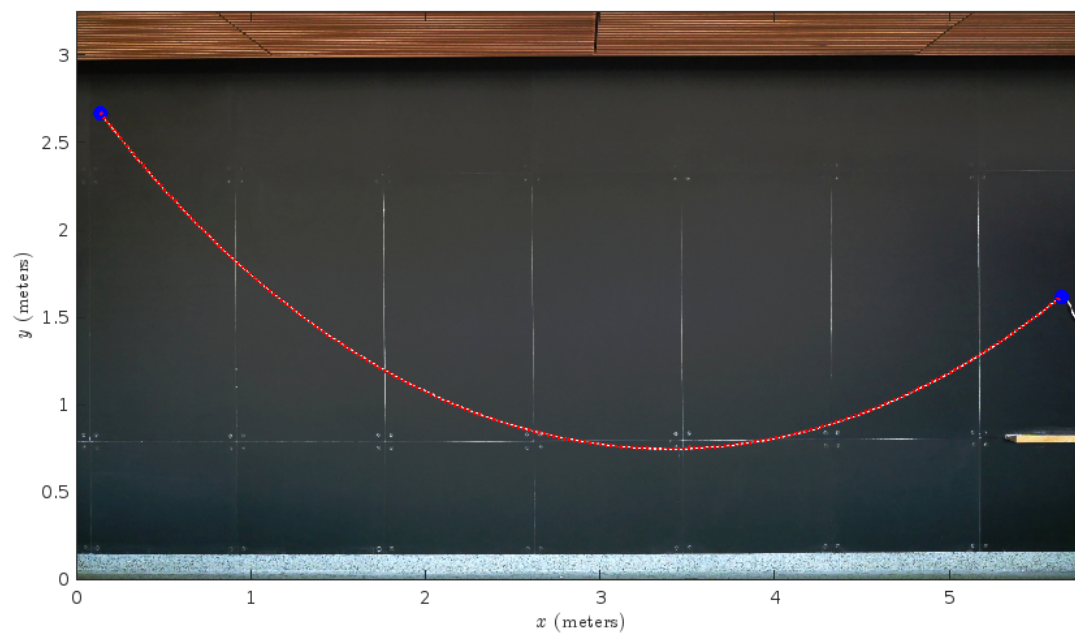
Assessment: All Tests Passed Submit ?

✔ cOut correct

Output

```
Equation solved (matlab: helpview('optim','eqn_solved','CSHelpWindow'));.

fsolve completed because the vector of function values is near zero
as measured by the value of the function tolerance (matlab: helpview('optim','fcn_tolerance_fsolve',
the problem appears regular (matlab: helpview('optim','appears_regular','CSHelpWindow')); as measure
<stopping criteria details (matlab: createExitMsg({'optim:fsolve:Exit1basic','fsolve'},{'optim:fsolv
```



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